Write a python program to illustrate ART neural network: Clustering 2D Data Using ART1-

a. Implement the ART1 algorithm for binary input data.

b. Given a set of binary vectors, cluster them into categories based on similarity.

c. Use vigilance parameter (ρ) to control cluster formation. Clustering 2D Data using Winner-Takes-All (WTA) Network Implement a Winner-Takes-All (WTA) neural network to cluster 2D data points.

d. Each neuron represents a cluster center, and only the winning neuron (closest to input) updates its weights.

e. Train the network on randomly generated 2D points and visualize the clusters.

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt

X=np.array([
    [1,1,0,0],
    [1,1,0,1],
    [0,0,1,1],
    [0,1,1,1],
    [1,0,0,0]
]
)
rho=0.75
w=[]

for x in X:
    assigned = False

    for i in range (len(w)):
        match=np.sum(np.minimum(x,w[i]))/np.sum(x)

        if match>=rho:
            w[i]=np.minimum(x,w[i])
            assigned=True
            break

    if not assigned:
        w.append(x.copy())
print("ART-I Clusters:")

for i,cluster in enumerate(w):
    print("Cluster",i+1,":",cluster)
```

```
    np.random.seed(42)

    data=np.vstack([
        np.random.randn(50,2)+[2,2],
        np.random.randn(50,2)+[7,7],
        np.random.randn(50,2)+[2,7]
    ])

k=3
alpha=0.1

centers=data[np.random.choice(len(data),k,replace=False)]

for epoch in range(50):
    for x in data:

        distance=np.linalg.norm(centers-x,axis=1)

        winner=np.argmin(distance)

        centers[winner]+=alpha*(x-centers[winner])

labels=[]

for x in data:
    distance=np.linalg.norm(centers-x,axis=1)
    labels.append(np.argmin(distance))

labels=np.array(labels)

print("\nWTA Cluster Centers: ")
print(centers)


plt.figure(figsize=(7,5))

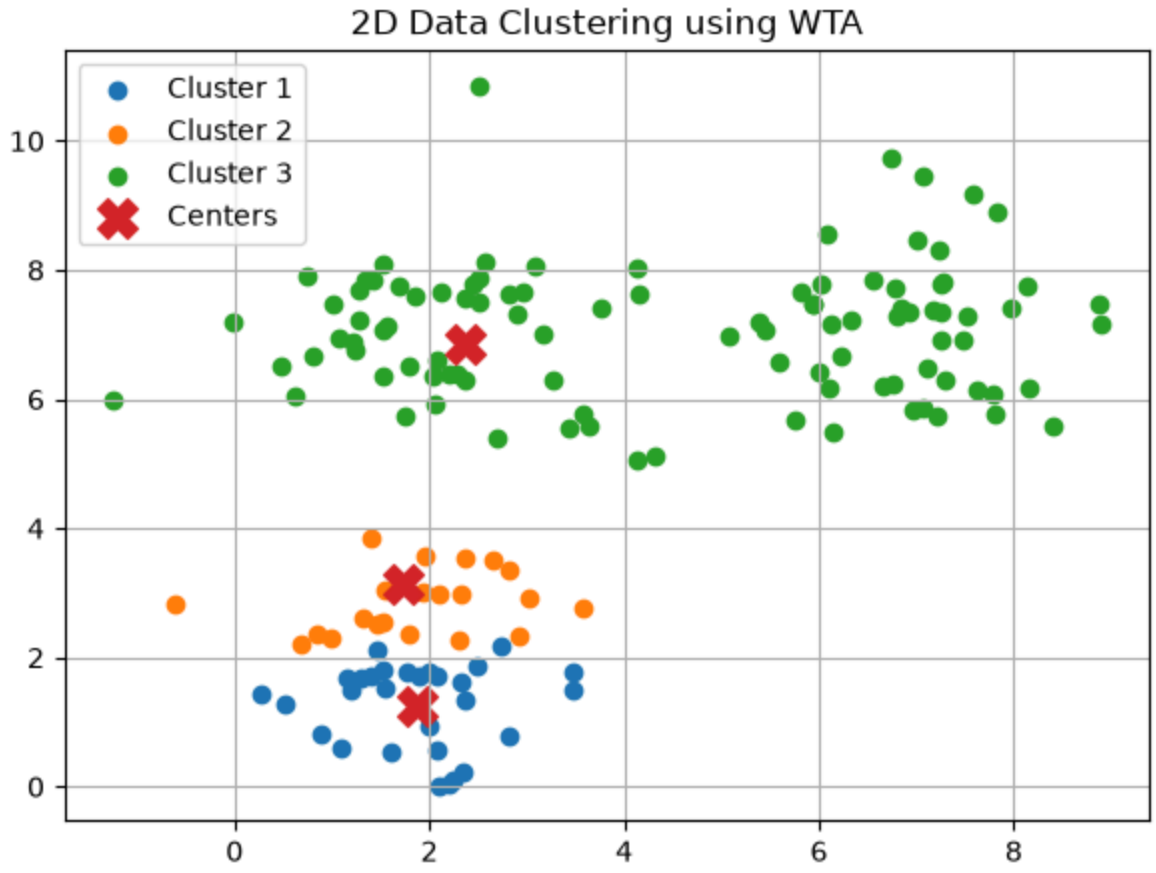
for i in range(k):
    points=data[labels==i]
    plt.scatter(points[:,0],points[:,1],
                label="Cluster "+ str(i+1))

plt.scatter(centers[:,0], centers[:,1],
           marker='X', s=200,label="Centers ")

plt.title("2D Data Clustering using WTA")
plt.legend()
plt.grid()
plt.show()
```

```
ART-I Clusters:
Cluster 1 : [1 0 0 0]
Cluster 2 : [1 1 0 1]
Cluster 3 : [0 0 1 1]
Cluster 4 : [0 1 1 1]

WTA Cluster Centers:
[[1.86809922 1.25553374]
 [1.73177976 3.13710984]
 [2.35806656 6.84714211]]
```



In [ ]: